

Veritas-Medica -- Market Readiness Roadmap

Date: 2026-08-03 **Horizon:** prototype -> first paying KSA hospital pilot -> commercial release **Factual base:** `PROTOTYPE_EVALUATION.md` and `STABILIZATION_AUDIT.md`. Nothing here restates code findings; this is the commercial and sequencing layer on top of them.

Method: RICE prioritisation with strategic overrides, Now/Next/Later sequencing (roadmap-planning skill), and a channel/messaging/metrics GTM structure (gtm-strategy skill). Frameworks are theirs; the facts, constraints and judgements are this codebase's.

0. The one thing that determines everything else

Three gates sit between this prototype and revenue, and none of them can be compressed by engineering effort:

Gate	Owner	Typical elapsed	Blocks
CCHI / NPHIES onboarding -- real IG profiles, conformance testing, certificates (annual renewal)	CCHI	Months, externally paced	Any real claim submission
SaMD-boundary sign-off for DDI + renal-dose + differential features already built	CTO + Clinical Advisor + Regulatory Consultant	Weeks-months	Any real-patient clinical use
SDAIA authorisation for cross-border health data, or in-Kingdom self-hosting	SDAIA / your infra	Weeks (self-host) vs uncertain (authorisation)	Any real patient record

Strategic consequence: the roadmap must be sequenced so engineering work runs *in parallel* with these gates, not behind them. The single highest-value technical move -- porting the orchestrator to the existing `ModelProvider` abstraction -- is valuable precisely because it **converts gate 3 from an external dependency into an internal one**. Self-hosting is something you control; SDAIA authorisation for a China-hosted LLM is not.

Corollary worth stating to leadership: a "market launch" date cannot be committed until gate 1 has a CCHI slot. Everything below is sequenced to make the company *ready* for that slot, and to generate revenue-adjacent proof in the meantime.

1. Corrections to the evaluation

Re-reading the evaluation with a commercial lens surfaces four things it under-weighted:

C-1 -- "Not pilot-ready" understates what *is* pilot-ready. The evaluation judges the whole product against real-patient use. But the **claim-integrity half** -- claim readiness, ICD/SBS coding, rejection-risk analytics, diagnosis linkage -- is administrative, not clinical. It touches no SaMD boundary and needs no clinical sign-off. It could pilot on real *claims* data far sooner than the clinical half can pilot on real *patients*. **This is the single biggest missed opportunity in the current framing** and it reshapes the roadmap (see §3, Now).

C-2 -- The competitive framing is too pessimistic on ambient, too modest on claims. Competing with Nuance DAX on ambient scribing is a losing fight. But no ambient vendor does NPHIES claim integrity, and no Saudi RCM vendor does graph-grounded explainability. The wedge is narrower and stronger than the evaluation states.

C-3 -- No pricing or commercial model exists anywhere in the documentation. Not a code gap, but you cannot go to market without one. Addressed in §4.

C-4 -- "515 tests pass" is an engineering metric, not a buyer metric. No clinical accuracy evaluation exists: no measured coding-suggestion precision, no false-positive rate on NPHIES flags, no scribe WER on Saudi-accented Arabic. A hospital CMO will ask for exactly these. This is a **gap the evaluation did not name** and is a Priority-1 addition.

2. Initiatives, scored

RICE over a 12-month horizon. *Reach* = share of the target pilot's daily workflow touched. *Impact* 0.5-3.

Confidence reflects both technical and external-dependency risk. *Effort* in person-weeks.

#	Initiative	R	I	C	E	RICE	Note
E1	Clinical accuracy evaluation pack (C-4)	3	3	90%	3	2.7	Unblocks every buyer conversation
E2	Port orchestrator to Model Provider + self-host demo	3	3	90%	1	8.1	Converts gate 3 to internal
E3	Claim-integrity pilot package (C-1)	2	3	80%	4	1.2	Revenue path that skips clinical gates
E4	Finish UI wiring (M-1, H-3b, M-7)	3	2	95%	2	2.9	Demo credibility
E5	Resolve ICD-10-A M vs CM + real IG profiles	3	3	50%	3	1.5	Externally gated; start discovery now
E6	Durable queues + multi-replica readiness	1	2	90%	3	0.6	Not needed for single-site pilot
E7	Arabic/dialect ASR improvement	2	2	60%	6	0.4	Expensive; validate demand first
E8	Neo4j PDPL retention/ensure parity	2	3	90%	2	2.7	Compliance blocker for real data

#	Initiative	R	I	C	E	RICE	Note
E9	ATC/therapeutic-class data for screening	1	2	70%	2	0.7	Fixes a credibility-damaging output
E10	SOC2 / HITRUST track	1	1	70%	12	0.06	Defer; not a KSA buying requirement

Strategic overrides applied:

- **E5 promoted** above its score. Its low confidence *is* the reason to start -- discovery is cheap and the answer gates everything downstream.
- **E8 promoted** to Now. It is small, and being unable to delete a patient from the graph is a PDPL conversation you cannot afford to lose mid-pilot.
- **E10 demoted** to Later. SOC2 matters for US enterprise; KSA buyers ask about NCA/SDAIA alignment instead. Do not spend 12 weeks on the wrong certification.
- **E7 demoted**. The evaluation flags dialect ASR as weak, but if the wedge is claims rather than ambient (C-2), this stops being on the critical path.

3. Roadmap -- Now / Next / Later

NOW -- next 6 weeks (committed)

Outcome: a demo that survives specialist scrutiny, and a defensible answer to "can you run this in our data centre?"

- 1 **E4 -- Finish the UI wiring.** Agent action cards (M-1), SOAP generation (H-3b), checklist persistence (M-7). Nothing else matters if a specialist clicks "Adjust Dosage" and nothing happens.
- 1 **E2 -- Port `deepseek_client.py` to the `ModelProvider` pattern** and demonstrate a self-hosted Llama 3 swap. ~1 week. Closes audit H-1 and the LLM lock-in simultaneously.
- 1 **E1 -- Build the clinical accuracy evaluation pack.** Coding-suggestion precision/recall against a labelled set, NPHIES flag false-positive rate, scribe WER on Arabic and English samples. **Publish the numbers even if they are unflattering** -- a vendor with measured weaknesses outranks one with no measurements.
- 1 **E8 -- Neo4j retention + erasure parity** with the existing Postgres DSR.
- 2 **E5 (discovery only)** -- get the ICD-10-AM/CM answer and request the real NPHIES IG. No code yet.

Exit criteria: demo runs end-to-end with no dead controls; a self-hosted model swap is demonstrable; an accuracy one-pager exists; erasure works.

NEXT -- 3 months (high confidence)

Outcome: a signed pilot on the half of the product that has no clinical gate.

- 1 **E3 -- Claim-integrity pilot package (C-1).** Package claim readiness, coding, rejection-risk analytics and pre-auth as a standalone offering that pilots on **real claims data with no real-patient clinical exposure**. This is your fastest credible path to a paying customer.
- 1 **E5 (execution) --** real IG profiles, conformance testing, certificate lifecycle. Flip **profiles_verified** to true only when genuinely true.
- 1 **E9 -- ATC/class data** so alternative screening stops proposing Warfarin as a Metformin alternative.
- 1 **SaMD sign-off process** started in parallel for the clinical half.
- 2 **Hijri dates, Arabic patient templates, clinician-gender scheduling --** small, visible, and repeatedly requested by Saudi clinicians.

Exit criteria: one hospital running claim integrity on real claims; NPHIES conformance passing; sign-off process underway.

LATER -- 6-12 months (lower confidence)

- 1 **Clinical pilot** once SaMD sign-off lands and in-Kingdom hosting is live.
- 2 **E6 -- Durable queues** when a second site or a second replica appears.
- 3 **E7 -- Dialect ASR**, only if pilot feedback shows ambient is the wedge.
- 4 **SMART-on-FHIR launch** for EMR embedding -- the main integration-breadth gap versus market leaders.
- 1 **E10 -- SOC2/HITRUST** only if pursuing non-KSA or large private groups.

4. Go-to-Market

Segment -- start narrow

Primary: mid-size private Saudi hospital groups (2-8 facilities) already submitting to NPHIES and feeling claim rejections in cash terms. They have budget authority, short decision chains, and a quantifiable pain. They are also small enough that a single-site, single-replica deployment is genuinely sufficient (which is why E6 can wait).

Not first: MOH facilities (procurement cycles measured in years) or single clinics (no budget, no NPHIES pain at scale).

Positioning

*For Saudi private hospital groups losing revenue to NPHIES claim rejections, **Veritas-Medica** is a clinical documentation and claim-integrity layer that validates every order against the documented diagnosis before submission -- and shows the exact graph path behind every decision. **Unlike** ambient scribes that stop at the note, or RCM tools that catch errors after rejection, it is grounded in a deterministic knowledge graph, so its claims are auditable rather than generated.*

Three proof points, in the order they land:

- 1 **"Click any claim and see the query."** Evidence chains are the demo moment

and the hardest thing for a competitor to retrofit.

1 **"It refuses rather than guesses."** The system declines to submit without a confirmed code, and only an explicit payer approval turns a badge green. In a market burned by AI overclaiming, restraint is a feature.

1 **"It runs in your data centre."** Post-E2, on your own hardware, with the PHI residency guard fail-closed by default.

Channels -- few, done well

Channel	Why	First action
Direct founder-led sales to hospital CFO/RCM directors	Claim rejection is a CFO-visible number	Build a 6-slide rejection-cost model per prospect
HIS/EMR vendor partnerships	They own the integration surface you lack	Target vendors already NPHIES-integrated
Clinical advisory board	Converts specialists into references	Recruit from the demo audience
Vision 2030 / health-transformation events	Where the buyers concentrate	One well-prepared demo beats a booth

Deliberately not: paid digital, content/SEO, product-led growth. Enterprise healthcare with a months-long cycle rewards none of them at this stage.

Commercial model (C-3 -- currently absent)

Recommended: **per-facility annual licence + per-clinician tier**, with a **value-linked pilot** -- priced against measured reduction in rejection rate, so the buyer's risk is bounded and your instrumentation becomes the sales asset. Avoid per-claim pricing: it caps your upside exactly when you succeed.

Metrics

Pilot success (the only ones that matter early):

- NPHIES rejection rate, pre- vs post-, on the same department
- Pre-auth turnaround time
- % of orders reaching submission with a confirmed code first time
- Clinician-minutes per encounter on documentation

Commercial: pilots started -> converted, ACV, cycle length, reference count.

Explicitly not: the -65% / -85% figures in the executive deck. Those are *targets*, and the pilot exists to replace them with measurements. Presenting them as achieved would repeat exactly the overclaiming this product is positioned against.

5. Recommended new features

Beyond fixing what exists -- ranked by buyer-visible value:

Feature	Rationale
Rejection-cost dashboard (SAR at risk, by department, by code)	Turns the product into a CFO conversation. Data already exists in the rejection-risk service.
Pre-submission claim simulator	"Run this claim before you send it" -- a natural, high-trust entry wedge.

Feature	Rationale
Coder review queue	Batch coding surface for RCM teams; today the flow is clinician-first only.
Accuracy transparency panel	Surface E1's measured precision/recall in-product. Differentiates hard against black-box vendors.
Audit export pack	One-click evidence bundle for CCHI/NCA review.
Ambient consent capture	Recording a consultation needs documented patient consent under PDPL. Currently absent and required before any real ambient use.

6. Immediate next actions

#	Action	Owner	When
1	Ask coding specialists: ICD-10-AM or CM?	Product	At the demo
2	Port orchestrator to ModelProvider ; demo self-hosted swap	Eng	Week 1-2
3	Wire M-1 / H-3b / M-7	Eng	Week 1-3
4	Build the accuracy evaluation pack	Eng + Clinical	Week 2-5
5	Start SaMD sign-off with Clinical Advisor + Regulatory	Leadership	This month
6	Request the real NPHIES IG; scope CCHI onboarding	Product	This month
7	Decide: claims-first wedge (C-1/C-2) or full-product?	Leadership	Before pilot outreach

Action 7 is the fork in the road. Claims-first reaches revenue months earlier because it sidesteps the SaMD and patient-data gates entirely. It also narrows the initial story. That is a leadership call, not an engineering one -- but the evaluation's own findings point clearly at claims-first.